1 Residual Helmholz Energy

$$\tilde{a}^{\text{res}} = \tilde{a}^{hc} + \tilde{a}^{\text{disp}} + \tilde{a}^{\text{assoc}} + \tilde{a}^{\text{DH}} + \tilde{a}^{\text{Born}} \tag{1}$$

1.1 Hard-Chain Reference Contribution

See [8].

$$\tilde{a}^{\text{hc}} = \bar{m}\tilde{a}^{\text{hs}} - \sum_i x_i \left(m_i - 1\right) \ln \mathrm{g}_{ii}^{\text{hs}} \left(\sigma_{ii}\right) \tag{2}$$

$$\bar{m} = \sum_i x_i m_i \tag{3}$$

$$\tilde{a}^{\text{hs}} = \frac{1}{\zeta_0} \left[\frac{3\zeta_1\zeta_2}{(1-\zeta_3)} + \frac{\zeta_2^3}{\zeta_3 \left(1-\zeta_3\right)^2} + \left(\frac{\zeta_2^3}{\zeta_3^2} - \zeta_0 \right) \ln\left(1-\zeta_3\right) \right] \tag{4}$$

$$\mathrm{g}_{ij}^{\text{hs}} = \frac{1}{(1-\zeta_3)} + \left(\frac{d_i d_j}{d_i + d_j} \right) \frac{3\zeta_2}{\left(1-\zeta_3\right)^2} + \left(\frac{d_i d_j}{d_i + d_j} \right)^2 \frac{2{\xi_2}^2}{\left(1-\xi_3\right)^3} \tag{5}$$

$$\xi_n = \frac{\pi}{6} \rho \sum_i x_i m_i d_i^n \quad n \in \{0, 1, 2, 3\} \tag{6}$$

$$d_i = \sigma_i \left[1 - 0.12 \exp\left(-3 \frac{\epsilon_i}{kT} \right) \right] \tag{7}$$

1.2 Dispersion Contribution

See [8].

$$\tilde{a}^{\text{disp}} = -2\pi\rho I_1(\eta, \bar{m}) \overline{m^2 \epsilon \sigma^3} - \pi\rho\bar{m} C_1 I_2(\eta, \bar{m}) \overline{m^2 \epsilon^2 \sigma^3} \tag{8}$$

$$C_1 = \left(1 + \bar{m}\frac{8\eta - 2\eta^2}{(1-\eta)^4} + \quad (1-\bar{m})\frac{20\eta - 27\eta^2 + 12\eta^3 - 2\eta^4}{[(1-\eta)(2-\eta)]^2}\right) \tag{9}$$

$$\overline{m^2\epsilon\sigma^3} = \sum_i \sum_j x_i x_j m_i m_j \left(\frac{\epsilon_{ij}}{kT}\right) \sigma_{ij}^3 \tag{10}$$

$$\overline{m^2\epsilon^2\sigma^3} = \sum_i \sum_j x_i x_j m_i m_j \left(\frac{\epsilon_{ij}}{kT}\right)^2 {\sigma_{ij}}^3 \tag{11}$$

$$\sigma_{ij} = \frac{1}{2}\left(\sigma_i + \sigma_j\right) \cdot \left(1 - l_{ij}\right) \tag{12}$$

$$\epsilon_{ij} = \begin{cases} \sqrt{\epsilon_i\,\epsilon_j}\left(1 - k_{ij}\right), & z_i z_j \neq 0, \\ \\ 0, & z_i z_j = 0. \end{cases} \tag{13}$$

$$\epsilon_{ij} = 0 \tag{14}$$

$$I_1(\eta, \bar{m}) = \sum_{i=0}^{6} a_i(\bar{m})\eta^i \tag{15}$$

$$I_2(\eta, \bar{m}) = \sum_{i=0}^{6} b_i(\bar{m})\eta^i \tag{16}$$

$$a_i(\bar{m}) = a_{0i} + \frac{\bar{m}-1}{\bar{m}}a_{1i} + \frac{\bar{m}-1}{\bar{m}}\frac{\bar{m}-2}{\bar{m}}a_{2i} \tag{17}$$

$$b_i(\bar{m}) = b_{0i} + \frac{\bar{m}-1}{\bar{m}}b_{1i} + \frac{\bar{m}-1}{\bar{m}}\frac{\bar{m}-2}{\bar{m}}b_{2i} \tag{18}$$

$$\rho = \frac{6}{\pi}\eta\left(\sum_i x_i m_i d_i^3\right)^{-1} \tag{19}$$

$$\hat{\rho} = \frac{\rho}{N_{\mathrm{AV}}}\left(10^{10}\frac{A^\circ}{\mathrm{m}}\right)^3\left(10^{-3}\frac{\mathrm{kmol}}{\mathrm{mol}}\right) \tag{20}$$

1.3 Association Contribution

See [6].

$$\tilde{a}^{\mathrm{assoc}} = \sum_i x_i \sum_{\mathrm{A}_i}\left(\ln X^{\mathrm{A}_i} - \frac{X^{\mathrm{A}_i}}{2} + \frac{1}{2}\right) \tag{21}$$

$$\begin{aligned} X^{\mathrm{A}_i} = &\left[1 + \sum_j \sum_{\mathrm{B}_j} \rho x_j X^{\mathrm{B}_j} \Delta^{\mathrm{A}_i\ \mathrm{B}_j}\right]^{-1}, \\ &\text{with } \sum_{\mathrm{B}_j} \text{ over all sites on molecule } j\ (\mathrm{A}_j, \mathrm{B}_j, \mathrm{C}_j, \ldots), \\ &\text{and } \sum_j \text{ over all components.} \end{aligned} \tag{22}$$

$$\rho_j = X_j \rho_{\mathrm{mixture}} \tag{23}$$

$$\Delta^{\mathrm{A},\ \mathrm{B}_j} = {d_{ij}}^3 \mathrm{g}_{ij}^{\mathrm{hs}}\left[\exp\left(\frac{\epsilon^{\mathrm{A},\ \mathrm{B}_j}}{kT}\right) - 1\right] \tag{24}$$

$$\varepsilon^{A_i B_j} = \frac{1}{2}(\varepsilon^{A_i B_i} + \varepsilon^{A_j B_j})(1 - k_{ij}^{\mathrm{hb}}) \tag{25}$$

$$k^{A_i B_j} = \sqrt{k^{A_i B_i} k^{A_j B_j}} \quad \left(\frac{\sqrt{\sigma_i \sigma_j}}{1/2(\sigma_i + \sigma_j)}\right)^3 \tag{26}$$

$$d_{ij} = (d_{ii} + d_{jj})/2. \tag{27}$$

1.4 Debye and Huckel Electrolyte Term Contribution

See [3].

$$\tilde{a}^{DH} = -\frac{\kappa e^2}{12\pi\varepsilon_0\varepsilon_r k_B T}\sum_i x_i z_i^2 \chi_i \tag{28}$$

$$\kappa = \sqrt{\frac{\rho e^2}{k_B T \varepsilon_0 \varepsilon_r}\sum_j x_j z_j^2} \tag{29}$$

$$\chi_i = \frac{3}{(\kappa d_i)^3}\left[\frac{3}{2} + \ln(1+\kappa d_i) - 2(1+\kappa d_i) + \frac{1}{2}(1+\kappa d_i)^2\right] \tag{30}$$

1.4.1 Ion Diameter

Temperature Independent Version

$$d_i = \sigma_i \qquad i \in \mathcal{I} \tag{31}$$

Temperature Dependent Version

$$d_i = [1 - 0.12]\,\sigma_i = .88\sigma_i \qquad i \in \mathcal{I} \tag{32}$$

Temperature Dependent Version 2

$$d_i = \left[1 - 0.12\exp\left(-3\frac{\epsilon_i}{kT}\right)\right]\sigma_i \qquad i \in \mathcal{I} \tag{33}$$

1.4.2 Dielectric Constant

See [1]. Mole-Fraction

$$\varepsilon_r = \sum_{j=1}^{N_c} \varepsilon_{r,j} x_j \tag{34}$$

Mass-Fraction

$$\varepsilon_r = \sum_{j=1}^{N_c} \varepsilon_{r,j}\, w_j = \frac{\sum_{j=1}^{N_c} \varepsilon_{r,j}\, x_j\, MW_j}{\sum_{j=1}^{N_c} x_j\, MW_j} = \frac{\sum_{j=1}^{N_c} \varepsilon_{r,j}\, x_j\, MW_j}{\overline{MW}} \tag{35}$$

Combo

$$\varepsilon_r = x_{sol}\, \varepsilon_{r,sol}^{(w)} + \sum_{j\in\mathcal{I}} \varepsilon_{r,j}\, x_j \tag{36}$$

$$\varepsilon_{r,sol}^{(w)} \equiv \sum_{j\in\mathcal{S}} \varepsilon_{r,j}\, w_j^{sol} = \frac{\sum_{j\in\mathcal{S}} \varepsilon_{r,j}\, x_j\, MW_j}{\sum_{j\in\mathcal{S}} x_j\, MW_j} = \frac{\sum_{j\in\mathcal{S}} \varepsilon_{r,j}\, x_j\, MW_j}{\overline{MW}_{sol}} \tag{37}$$

New Mixing Rule (2025) [7]

$$\varepsilon_{r,\text{mix}} = \frac{\varepsilon_{r,\text{solvent,mix}}^{\text{salt-free}}}{1 + 7.01 \cdot x_{\text{ion}}} \tag{38}$$

$$\varepsilon_{r,\text{solvent,mix}}^{\text{salt-free}} = \sum_{\text{solvent}} w_{\text{solvent}}^{\text{salt-free}} \cdot \varepsilon_{r,\text{solvent}} \tag{39}$$

Intermediates

$$\overline{MW} = \sum_{j=1}^{N_c} x_j\, MW_j \tag{40}$$

$$\overline{MW}_{sol} = \sum_{j\in\mathcal{S}} x_j\, MW_j \tag{41}$$

$$\mathcal{S} = \{j : z_j = 0\}, \qquad \mathcal{I} = \{j : z_j \neq 0\} \tag{42}$$

$$x_{sol} = \sum_{j \in \mathcal{S}} x_j \tag{43}$$

1.4.3 Bjerrum Treatment

See [2].

$$\tilde{a}^{DH} = -\frac{\kappa e^2}{12\pi\varepsilon_0\varepsilon_r k_B T} \sum_i \alpha_i x_i z_i^2 \chi_i \tag{44}$$

$$\kappa = \sqrt{\frac{\rho e^2}{k_B T \varepsilon_0 \varepsilon_r} \sum_j \alpha_j x_j z_j^2} \tag{45}$$

$$\chi_i = \frac{3}{(\kappa R_i)^3}\left[\frac{3}{2} + \ln(1+\kappa R_i) - 2(1+\kappa R_i) + \frac{1}{2}(1+\kappa R_i)^2\right] \tag{46}$$

Bjerrum Length

$$R_i = \begin{cases} d_{ion}, & d_{ion} > l_B, \\ l_B, & d_{ion} < l_B. \end{cases} \tag{47}$$

$$l_B = \frac{|z_i z_j|\, e^2}{8\pi\varepsilon_0\varepsilon_r k_B T} \tag{48}$$

Dissociation Degree Solved implicitly

$$\alpha = \frac{-1 + \sqrt{1 + 4x_{\pm} K_{ip} \frac{\left(\gamma_{\pm}^{*}(x_f)\right)^2}{\gamma_{ip}^{*}}}}{2x_{\pm} K_{ip} \frac{\left(\gamma_{\pm}^{*}(x_f)\right)^2}{\gamma_{ip}^{*}}} \tag{49}$$

$$K_{ip}(T)=\frac{(1-\alpha)\cdot\gamma_{ip}^{*}}{\alpha^{2}\cdot x_{\pm}\cdot\left(\gamma_{\pm}^{*}(x_{f}(\alpha))\right)^{2}}=4\pi\rho_{N}\int_{a}^{l_{B}}\exp\left(\frac{|z_{i}z_{j}|\,e^{2}}{4\pi\varepsilon_{0}\varepsilon_{r}k_{B}T}\cdot\frac{1}{r}\right)r^{2}dr \tag{50}$$

$$\gamma_{ip}^{*}\approx 1 \tag{51}$$

$$x_{\pm}=x_{f}+x_{ip} \tag{52}$$

$$x_{\pm}=\alpha x_{f} \tag{53}$$

$$x_{\pm}=(1-\alpha)x_{ip} \tag{54}$$

$$x_{\pm}=\left(x_{c}^{\nu c}\cdot x_{a}^{\nu a}\right)^{\frac{1}{\nu_{c}+\nu_{a}}} \tag{55}$$

$$\mu_{i}^{DH}\left(x_{f}\right)=\left(\frac{\partial A^{DH}}{\partial\rho_{i}\left(x_{f}\right)}\right)_{T,V,N_{j\neq i}}=-\frac{e^{2}z_{i}^{2}\kappa}{24\pi\varepsilon_{0}\varepsilon_{r}}\left[2\chi_{i}+\frac{\sum_{k}x_{k,f}z_{k}^{2}\sigma_{k}}{\sum_{k}x_{k,f}z_{k}^{2}}\right] \tag{56}$$

$$\ln\gamma_{i}^{*}\left(x_{f,i}\right)=-\frac{e^{2}z_{i}^{2}\kappa}{24\pi\varepsilon_{0}\varepsilon_{r}}\left[2\chi_{i}+\frac{\sum_{k}x_{k,f}z_{k}^{2}\sigma_{k}}{\sum_{k}x_{k,f}z_{k}^{2}}\right] \tag{57}$$

$$\gamma_{\pm}^{*}=\left(\gamma_{c}^{*,\nu c}\cdot\gamma_{a}^{*,\nu a}\right)^{\frac{1}{\nu_{c}+\nu_{a}}} \tag{58}$$

1.5 Born Electrolyte Term Contribution

See [4].

$$\tilde{a}^{\text{Born}}\left(\varepsilon(x)\right) = -\frac{e^2}{4\pi\varepsilon_0 k_B T}\left(1-\frac{1}{\varepsilon_{r,\text{bulk}}}\right)\sum_i \frac{x_i z_i^2}{d_i^{\text{Born}}} \tag{59}$$

$$d_i^{\text{Born}} = d_i \qquad i \in \mathcal{I} \tag{60}$$

1.5.1 Solvation Shell Model + Dielectric Saturation

Version 1a: Equation taken directly from the paper (probably wrong): [7]

$$\begin{aligned} a_{\text{SSM+DS}}^{\text{Born}} = &-\frac{e^2}{4\pi\varepsilon_0 k_{\text{B}} T}\left(1-\frac{1}{\varepsilon_{r,\text{bulk}}}\right)\sum_i \frac{x_i z_i^2}{d_i^{\text{Born}}+\Delta d_i} \\ &+\left(1-\frac{1}{\varepsilon_{r,\text{bulk}}}\right)\left(\sum_i x_i z_i^2\left(\frac{1}{d_i^{\text{Born}}}-\frac{1}{d_i^{\text{Born}}+\Delta d_i}\right)\right) \end{aligned} \tag{61}$$

Version 1b: Equation from paper with modification to include the first fraction for the DS term (probably wrong): [7]

$$\begin{aligned} a_{\text{SSM+DS}}^{\text{Born}} = &-\frac{e^2}{4\pi\varepsilon_0 k_{\text{B}} T}\left(1-\frac{1}{\varepsilon_{r,\text{bulk}}}\right)\sum_i \frac{x_i z_i^2}{d_i^{\text{Born}}+\Delta d_i} \\ &-\frac{e^2}{4\pi\varepsilon_0 k_{\text{B}} T}\left(1-\frac{1}{\varepsilon_{r,\text{bulk}}}\right)\left(\sum_i x_i z_i^2\left(\frac{1}{d_i^{\text{Born}}}-\frac{1}{d_i^{\text{Born}}+\Delta d_i}\right)\right) \end{aligned} \tag{62}$$

Version 2 (my version): potential fix (changing the second $\varepsilon_{r,\text{solv}}$ to $\varepsilon_{r,\text{ion}}$ and including the first fraction for the DS term)

$$a_{\text{SSM}}^{\text{Born}} = -\frac{e^2}{4\pi\varepsilon_0 k_{\text{B}} T}\sum_i x_i z_i^2\left(1-\frac{1}{\varepsilon_{r,\text{bulk}}}\right)\left(\frac{1}{d_i^{\text{Born}}+\Delta d_i}\right) \tag{63}$$

$$a_{\text{DS}}^{\text{Born}} = -\frac{e^2}{4\pi\varepsilon_0 k_{\text{B}} T}\sum_i x_i z_i^2\left(1-\frac{1}{\varepsilon_{r,\text{ion}}}\right)\left(\frac{1}{d_i^{\text{Born}}}-\frac{1}{d_i^{\text{Born}}+\Delta d_i}\right) \tag{64}$$

$$
\begin{aligned}
a^{\mathrm{Born}}_{\mathrm{SSM+DS}} = &-\frac{e^2}{4\pi\varepsilon_0 k_{\mathrm{B}} T}\sum_i x_i z_i^2 \left(1-\frac{1}{\varepsilon_{r,\mathrm{bulk}}}\right)\left(\frac{1}{d_i^{\mathrm{Born}}+\Delta d_i}\right) \\
&-\frac{e^2}{4\pi\varepsilon_0 k_{\mathrm{B}} T}\sum_i x_i z_i^2 \left(1-\frac{1}{\varepsilon_{r,\mathrm{ion}}}\right)\left(\frac{1}{d_i^{\mathrm{Born}}}-\frac{1}{d_i^{\mathrm{Born}}+\Delta d_i}\right)
\end{aligned} \tag{65}
$$

$$
\Delta d_i = \frac{(f_{mix}-1)}{|z_i|}\cdot d_i^{\mathrm{Born}} \tag{66}
$$

$$
f_{mix} = \sum_k x_k f_k \tag{67}
$$

$$
d_i^{\mathrm{Born}} = d_i^{\mathrm{Born,\ fitted}} \qquad i \in \mathcal{I} \tag{68}
$$

2 Pressure (Compressibility Factor)

See [8].

$$
P = ZkT\rho\left(10^{10}\frac{\hat{\mathrm{A}}}{\mathrm{m}}\right)^3 \tag{69}
$$

$$
Z = 1+\eta\left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial \eta}\right)_{T,x_i} \tag{70}
$$

$$
\eta\left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial \eta}\right)_{T,x_i} = \rho\left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial \rho}\right)_{T,x_i} \tag{71}
$$

$$
Z = 1+\rho\left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial \rho}\right)_{T,x_i} \tag{72}
$$

$$
Z = 1 + Z^{hc} + Z^{disp} + Z^{assoc} + Z^{DH} + Z^{Born} \tag{73}
$$

$$Z - 1 = Z^{hc} + Z^{disp} + Z^{assoc} + Z^{DH} + Z^{Born} \tag{74}$$

$$Z^{\alpha} = \rho \left(\frac{\partial \tilde{a}^{\alpha}}{\partial \rho} \right)_{T,x} . \tag{75}$$

2.1 Hard-Chain Reference Contribution

See [8].

$$Z^{hc} = \rho \left(\frac{\partial \tilde{a}^{hc}}{\partial \rho} \right)_{T,x} . \tag{76}$$

$$Z^{\text{hc}} = \bar{m} Z^{\text{hs}} - \sum_i x_{\text{i}} (m_i - 1)(g_{ii}^{\text{hs}})^{-1} \rho \frac{\partial g_{ii}^{\text{hs}}}{\partial \rho} \tag{77}$$

$$Z^{hs} = \rho \left(\frac{\partial \tilde{a}^{hs}}{\partial \rho} \right)_{T,x} . \tag{78}$$

$$Z^{\text{hs}} = \frac{\zeta_3}{(1 - \zeta_3)} + \frac{3\zeta_1\zeta_2}{\zeta_0(1-\zeta_3)^2} + \frac{3\zeta_2^3 - \zeta_3\zeta_2^3}{\zeta_0(1-\zeta_3)^3} \tag{79}$$

$$\begin{aligned} \rho \frac{\partial g_{ij}^{\text{hs}}}{\partial \rho} = \frac{\zeta_3}{(1-\zeta_3)^2} &+ \left(\frac{d_i d_j}{d_i + d_j} \right) \left(\frac{3\zeta_2}{(1-\zeta_3)^2} + \frac{6\zeta_2\zeta_3}{(1-\zeta_3)^3} \right) \\ &+ \left(\frac{d_i d_j}{d_i + d_j} \right)^2 \left(\frac{4{\zeta_2}^2}{(1-\zeta_3)^3} + \frac{6{\zeta_2}^2\zeta_3}{(1-\zeta_3)^4} \right) \end{aligned} \tag{80}$$

2.2 Dispersion Contribution

See [8].

$$Z^{disp} = \rho \left(\frac{\partial \tilde{a}^{disp}}{\partial \rho} \right)_{T,x} . \tag{81}$$

$$
\begin{aligned}
Z^{\text{disp}} = &-2\pi\rho\frac{\partial(\eta I_1)}{\partial\eta}\overline{m^2\epsilon\sigma^3} \\
&-\pi\rho\bar{m}\left[C_1\frac{\partial(\eta I_2)}{\partial\eta}+C_2\eta I_2\right]\overline{m^2\epsilon^2\sigma^3}
\end{aligned}
\tag{82}
$$

$$
\frac{\partial(\eta I_1)}{\partial\eta}=\sum_{j=0}^{6}a_j(\bar{m})(j+1)\eta^j \tag{83}
$$

$$
\frac{\partial(\eta I_2)}{\partial\eta}=\sum_{i=0}^{6}b_j(\bar{m})(j+1)\eta^i \tag{84}
$$

$$
C_2=\frac{\partial C_1}{\partial\eta}=-C_1^2\left(\bar{m}\frac{-4\eta^2+20\eta+8}{(1-\eta)^5}+(1-\bar{m})\frac{2\eta^3+12\eta^2-48\eta+40}{[(1-\eta)(2-\eta)]^3}\right) \tag{85}
$$

2.3 Association Contribution

See [6].

$$
Z^{assoc}=\rho\left(\frac{\partial\tilde{a}^{assoc}}{\partial\rho}\right)_{T,x}. \tag{86}
$$

$$
Z^{assoc}=\rho\sum_i x_i\sum_{A_i}\left(\frac{1}{X^{A_i}}-\frac{1}{2}\right)\left(\frac{\partial X^{A_i}}{\partial\rho}\right)_{T,x}. \tag{87}
$$

$$
\begin{aligned}
\left(\frac{\partial X^{A_i}}{\partial\rho}\right)_{T,x} = &-(X^{A_i})^2\sum_j\sum_{B_j}x_j\Bigg[X^{B_j}\Delta^{A_iB_j} \\
&+\rho\left(\Delta^{A_iB_j}\left(\frac{\partial X^{B_j}}{\partial\rho}\right)_{T,x}+X^{B_j}\left(\frac{\partial\Delta^{A_iB_j}}{\partial\rho}\right)_{T,x}\right)\Bigg]
\end{aligned}
\tag{88}
$$

$$
\left(\frac{\partial\Delta^{A_iB_j}}{\partial\rho}\right)_{T,x}=d_{ij}^3\kappa^{A_iB_j}\left[\exp(\epsilon^{A_iB_j}/kT)-1\right]\left(\frac{\partial g_{ij}^{hs}}{\partial\rho}\right)_{T,x} \tag{89}
$$

2.4 Debye and Huckel Electrolyte Term Contribution

$$Z^{DH} = \rho \left(\frac{\partial \tilde{a}^{DH}}{\partial \rho} \right)_{T,x}. \tag{90}$$

See [5].

$$Z^{DH} = -\frac{\kappa e^2}{24\pi kT\epsilon} \sum_i x_i {z_i}^2 \sigma_i \tag{91}$$

$$\sigma_i = \left(\frac{\partial(\kappa\chi_i)}{\partial\kappa} \right)_{T,\mathrm{x}} = -2\chi_i + \frac{3}{1+\kappa a_i} \tag{92}$$

2.4.1 Bjerrum Treatment

See [2].

$$Z^{DH} = -\frac{\kappa e^2}{24\pi kT\epsilon} \sum_i \alpha_i x_i {z_i}^2 \sigma_i \tag{93}$$

$$\sigma_i = \left(\frac{\partial(\kappa\chi_i)}{\partial\kappa} \right)_{T,\mathrm{x}} = -2\chi_i + \frac{3}{1+\kappa R_i} \tag{94}$$

2.5 Born Electrolyte Term Contribution

$$Z^{Born} = \rho \left(\frac{\partial \tilde{a}^{Born}}{\partial \rho} \right)_{T,x}. \tag{95}$$

$$Z^{Born} = 0 \tag{96}$$

3 Chemical Potential

See [8].

$$\tilde{\mu_k}^{\text{res}} = \frac{\mu_k^{\text{res}}}{kT} = \frac{\hat{\mu}_k^{\text{res}}}{RT} \tag{97}$$

$$\begin{aligned}\tilde{\mu_k}^{\text{res}} = \tilde{a}^{\text{res}} + (\mathrm{Z}-1) + \left(\frac{\partial \tilde{a}^{\text{res}}}{\partial x_k}\right)_{T,\nu,x_{i\neq k}} \\ - \sum_{j=1}^{N}\left[x_j\left(\frac{\partial \tilde{a}^{\text{res}}}{\partial x_j}\right)_{T,\nu,x_{i\neq j}}\right]\end{aligned} \tag{98}$$

$$\tilde{\mu_k}^{\text{res}} = \tilde{\mu_k}^{\text{hc}} + \tilde{\mu_k}^{\text{disp}} + \tilde{\mu_k}^{\text{assoc}} + \tilde{\mu_k}^{\text{DH}} + \tilde{\mu_k}^{\text{Born}} \tag{99}$$

$$\zeta_{n,xk} = \left(\frac{\partial \zeta_n}{\partial x_k}\right)_{T,\boldsymbol{\rho},x_{j\neq k}} = \frac{\pi}{6}\rho m_k (d_k)^n \quad n \in \{0,1,2,3\} \tag{100}$$

3.1 Hard-Chain Reference Contribution

See [8].

$$\begin{aligned}\tilde{\mu}_k^{hc} = \tilde{a}^{\text{hc}} + Z^{hc} + \left(\frac{\partial \tilde{a}^{\text{hc}}}{\partial x_k}\right)_{T,\nu,x_{i\neq k}} \\ - \sum_{j=1}^{N}\left[x_j\left(\frac{\partial \tilde{a}^{\text{hc}}}{\partial x_j}\right)_{T,\nu,x_{i\neq j}}\right]\end{aligned} \tag{101}$$

$$\begin{aligned}\left(\frac{\partial \tilde{a}^{\text{hc}}}{\partial x_k}\right)_{T,\boldsymbol{\rho},x_{j\neq k}} &= m_k \tilde{a}^{\text{hs}} + \bar{m}\left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial x_k}\right)_{T,\boldsymbol{\rho},x_{j\neq k}} \\ &\quad - \sum_i x_i (m_i - 1)(g_{ii}^{\text{hs}})^{-1}\left(\frac{\partial g_{ii}^{\text{hs}}}{\partial x_k}\right)_{T,\boldsymbol{\rho},x_{j\neq k}}\end{aligned} \tag{102}$$

$$
\begin{aligned}
\left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial x_k}\right)_{T,\boldsymbol{\rho},x_{j\neq k}} = & -\frac{\zeta_{0,xk}}{\zeta_0}\tilde{a}^{\text{hs}} + \frac{1}{\zeta_0}\left[\frac{3(\zeta_{1,xk}\zeta_2+\zeta_1\zeta_{2,xk})}{(1-\zeta_3)} + \frac{3\zeta_1\zeta_2\zeta_{3,xk}}{(1-\zeta_3)^2}\right.\\
& + \frac{3\zeta_2^2\zeta_{2,xk}}{\zeta_3(1-\zeta_3)^2} + \frac{\zeta_2^3\zeta_{3,xk}(3\zeta_3-1)}{\zeta_3^2(1-\zeta_3)^3}\\
& + \left(\frac{3{\zeta_2}^2\zeta_{2,xk}\zeta_3 - 2{\zeta_2}^3\zeta_{3,xk}}{{\zeta_3}^3} - \zeta_{0,xk}\right)\ln(1-\zeta_3)\\
& \left. + \left(\zeta_0 - \frac{\zeta_2^3}{\zeta_3^2}\right)\frac{\zeta_{3,xk}}{(1-\zeta_3)}\right]
\end{aligned} \tag{103}
$$

$$
\begin{aligned}
\left(\frac{\partial g_{ij}^{\text{hs}}}{\partial x_k}\right)_{T,\boldsymbol{\rho},\boldsymbol{\rho}_{j\neq k}} = & \frac{\zeta_{3,x,k}}{(1-\zeta_3)^2} + \left(\frac{d_i d_j}{d_i+d_j}\right)\left(\frac{3\zeta_{2,x,k}}{(1-\zeta_3)^2} + \frac{6\zeta_2\zeta_{3,x,k}}{(1-\zeta_3)^3}\right)\\
& + \left(\frac{d_i d_j}{d_i+d_j}\right)^2\left(\frac{4\zeta_2\zeta_{2,x,k}}{(1-\zeta_3)^3} + \frac{6\zeta_2^2\zeta_{3,x,k}}{(1-\zeta_3)^4}\right)
\end{aligned} \tag{104}
$$

3.2 Dispersion Contribution

See [8].

$$
\begin{aligned}
\tilde{\mu}_k^{disp} = \tilde{a}^{\text{disp}} + Z^{disp} + \left(\frac{\partial \tilde{a}^{\text{disp}}}{\partial x_k}\right)_{T,\nu,x_{i\neq k}}\\
- \sum_{j=1}^{N}\left[x_j\left(\frac{\partial \tilde{a}^{\text{disp}}}{\partial x_j}\right)_{T,\nu,x_{i\neq j}}\right]
\end{aligned} \tag{105}
$$

$$
\begin{aligned}
\left(\frac{\partial \tilde{a}^{\text{disp}}}{\partial x_k}\right)_{T,\boldsymbol{p},x_{j\neq k}} = & -2\pi\rho\left[I_{1,xk}\overline{m^2\epsilon\sigma^3} + I_1\overline{(m^2\epsilon\sigma^3)_{xk}}\right]\\
& -\pi\rho\left\{\left[m_k C_1 I_2 + \bar{m}C_{1,xk}I_2 + \bar{m}C_1 I_{2,xk}\right]\overline{m^2\epsilon^2\sigma^3} + \bar{m}C_1 I_2\left(\bar{m}^2\epsilon^2\sigma^3\right)_{xk}\right\}
\end{aligned} \tag{106}
$$

$$
\overline{(m^2\epsilon\sigma^3)}_{xk} = 2m_k\sum_j x_j m_j\left(\frac{\epsilon_{kj}}{kT}\right)\sigma_{kj}^3 \tag{107}
$$

$$
(\overline{m^2\epsilon^2\sigma^3})_{xk} = 2m_k\sum_j x_j m_j\left(\frac{\epsilon_{kj}}{kT}\right)^2\sigma_{kj}^3 \tag{108}
$$

$$C_{1,xk} = C_2\zeta_{3,xk} - C_1^2 \left\{ m_k \frac{8\eta - 2\eta^2}{(1-\eta)^4} - m_k \frac{20\eta - 27\eta^2 + 12\eta^3 - 2\eta^4}{[(1-\eta)(2-\eta)]^2} \right\} \tag{109}$$

$$I_{1,xk} = \sum_{i=0}^{6} [a_{\mathrm{i}}(\bar{m}) i \zeta_{3,xk} \eta^{i-1} + a_{\mathrm{i,xk}} \eta^i] \tag{110}$$

$$I_{2,xk} = \sum_{i=0}^{6} [b_{\mathrm{i}}(\bar{m}) i \zeta_{3,xk} \eta^{i-1} + b_{\mathrm{i,xk}} \eta^i] \tag{111}$$

$$a_{\mathrm{i},xk} = \frac{m_k}{\bar{m}^2} a_{1i} + \frac{m_k}{\bar{m}^2} \left(3 - \frac{4}{\bar{m}}\right) a_{2i} \tag{112}$$

$$b_{\mathrm{i},xk} = \frac{m_k}{\bar{m}^2} b_{1i} + \frac{m_k}{\bar{m}^2} \left(3 - \frac{4}{\bar{m}}\right) b_{2i} \tag{113}$$

3.3 Association Contribution

See [6].

$$\begin{aligned} \tilde{\mu}_k^{assoc} = \tilde{a}^{\mathrm{assoc}} + Z^{assoc} + \left(\frac{\partial \tilde{a}^{\mathrm{assoc}}}{\partial x_k}\right)_{T,\nu,x_{i\neq k}} \\ - \sum_{j=1}^{N} \left[x_j \left(\frac{\partial \tilde{a}^{\mathrm{assoc}}}{\partial x_j}\right)_{T,\nu,x_{i\neq j}} \right] \end{aligned} \tag{114}$$

$$\left(\frac{\partial \tilde{a}^{assoc}}{\partial x_k}\right)_{T,v,x_{i\neq k}} = \sum_{A_k} \left(\ln X^{A_k} - \frac{X^{A_k}}{2} + \frac{1}{2} \right) + \sum_i x_i \sum_{A_i} \left(\frac{1}{X^{A_i}} - \frac{1}{2} \right) \left(\frac{\partial X^{A_i}}{\partial x_k} \right)_{T,v,x_{i\neq k}} \tag{115}$$

$$
\begin{aligned}
\left(\frac{\partial X^{A_i}}{\partial x_k}\right)_{T,v,x_{j\neq k}} &= -(X^{A_i})^2\,\rho\Bigg[\sum_{B_k} X^{B_k}\Delta^{A_iB_k} \\
&\quad + \sum_j x_j \sum_{B_j}\left(\Delta^{A_iB_j}\left(\frac{\partial X^{B_j}}{\partial x_k}\right)_{T,v,x_{j\neq k}} + X^{B_j}\left(\frac{\partial \Delta^{A_iB_j}}{\partial x_k}\right)_{T,v,x_{j\neq k}}\right)\Bigg]
\end{aligned} \tag{116}
$$

$$
\left(\frac{\partial \Delta^{A_iB_j}}{\partial x_k}\right)_{T,v,x_{j\neq k}} = d_{ij}^3\kappa^{A_iB_j}\left[\exp\left(\frac{\epsilon^{A_iB_j}}{kT}\right) - 1\right]\left(\frac{\partial g_{ij}^{hs}}{\partial x_k}\right)_{T,\rho,x_{j\neq k}} \tag{117}
$$

3.4 Debye and Huckel Electrolyte Term Contribution

$$
\begin{aligned}
\tilde{\mu}_k^{DH} &= \tilde{a}^{\mathrm{DH}} + Z^{DH} + \left(\frac{\partial \tilde{a}^{\mathrm{DH}}}{\partial x_k}\right)_{T,\nu,x_{i\neq k}} \\
&\quad - \sum_{j=1}^{N}\left[x_j\left(\frac{\partial \tilde{a}^{\mathrm{DH}}}{\partial x_j}\right)_{T,\nu,x_{i\neq j}}\right]
\end{aligned} \tag{118}
$$

The original 2005 version from [5]

$$
\tilde{\mu}_i^{DH} = -\frac{q_i^2\kappa}{24\pi kT\epsilon}\left[2\chi_i + \frac{\sum_j x_j q_j^2\sigma_k}{\sum_j x_j q_j^2}\right] \tag{119}
$$

$$
\sigma_k = \left(\frac{\partial\left(\kappa\chi_k\right)}{\partial\kappa}\right)_{T,\ \mathrm{N}} = -2\chi_k + \frac{3}{1+\kappa d_k} \tag{120}
$$

The updated version for concentration dependent dielectric constant from [3]

$$
\begin{aligned}
\left(\frac{\partial \tilde{a}^{DH}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = -\frac{e^2}{12\pi\varepsilon_0 k_B T}\Bigg[& \frac{1}{\varepsilon_r}\left(\frac{\partial \kappa}{\partial x_i}\right)_{T,v,x_{j\neq i}} \sum_j x_j z_j^2 \chi_j \\
& -\frac{\kappa}{\varepsilon_r^2}\left(\frac{\partial \varepsilon_r}{\partial x_i}\right)_{T,v,x_{j\neq i}} \sum_j x_j z_j^2 \chi_j \\
& +\frac{\kappa}{\varepsilon_r}\sum_j \chi_j z_j^2 \left(\frac{\partial x_j}{\partial x_i}\right)_{T,v,x_{k\neq i}} \\
& +\frac{\kappa}{\varepsilon_r}\sum_j x_j z_j^2 \left(\frac{\partial \chi_j}{\partial x_i}\right)_{T,v,x_{k\neq i}} \Bigg]
\end{aligned}
\tag{121}
$$

$$
\begin{aligned}
& \left(\frac{\partial \kappa}{\partial x_i}\right)_{T,v,x_{j\neq i}} = \frac{1}{2}\left(\frac{\rho_N e^2}{k_B T \varepsilon_0 \varepsilon_r}\sum_j x_j z_j^2\right)^{-\frac{1}{2}} \\
& \left\{-\frac{\rho_N e^2}{k_B T\left(\varepsilon_0\varepsilon_r\right)^2}\varepsilon_0 \frac{\partial\left(\varepsilon_r\right)}{\partial x_i}\sum_j x_j z_j^2 + \frac{\rho_N e^2}{k_B T \varepsilon_0 \varepsilon_r}\alpha_i z_i^2\right\} \\
& = \left(\frac{\rho_N e^2}{k_B T}\right)^{\frac{1}{2}} \left\{ -\frac{1}{2}\left(\varepsilon_0\varepsilon_r\right)^{-\frac{3}{2}}\varepsilon_0 \frac{\partial\left(\varepsilon_r\right)}{\partial x_i}\left[\sum_j x_j z_j^2\right]^{\frac{1}{2}} \right. \\
& \left. + \frac{1}{2\sqrt{\varepsilon_0\varepsilon_r}}\left[\sum_j x_j z_j^2\right]^{-\frac{1}{2}} \alpha_i z_i^2 \right\}
\end{aligned}
\tag{122}
$$

$$
\begin{aligned}
\left(\frac{\partial \chi_i}{\partial x_i}\right)_{T,v,x_{j\neq i}} & = -\frac{9}{\left(\kappa d_i\right)^4}\left[\frac{3}{2} + \ln(1+\kappa d_i) - 2(1+\kappa d_i) \right. \\
& \left. + \frac{1}{2}\left(1+\kappa d_i\right)^2\right] d_i \frac{\partial \kappa}{\partial x_i} + \frac{3}{\left(\kappa d_i\right)^3}\left[\frac{1}{1+\kappa d_i} - 1 + \kappa d_i\right] \\
d_i \frac{\partial \kappa}{\partial x_i} & = 3\frac{\partial \kappa}{\partial x_i}\left\{-\frac{\chi_i}{\kappa} + \frac{d_i}{\left(\kappa d_i\right)^3}\left[\frac{1}{1+\kappa d_i} - 1 + \kappa d_i\right]\right\}
\end{aligned}
\tag{123}
$$

3.4.1 Bjerrum Treatment

See [2].

$$\left(\frac{\partial \tilde{a}^{DH}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = -\frac{e^2}{12\pi\varepsilon_0 k_B T}\left[\frac{1}{\varepsilon_r}\left(\frac{\partial \kappa}{\partial x_i}\right)_{T,v,x_{j\neq i}}\sum_j \alpha_j x_j z_j^2 \chi_j \right.$$
$$-\frac{\kappa}{\varepsilon_r^2}\left(\frac{\partial \varepsilon_r}{\partial x_i}\right)_{T,v,x_{j\neq i}}\sum_j \alpha_j x_j z_j^2 \chi_j$$
$$+\frac{\kappa}{\varepsilon_r}\alpha_i z_i^2 \chi_i$$
$$\left. +\frac{\kappa}{\varepsilon_r}\sum_j \alpha_j x_j z_j^2 \left(\frac{\partial \chi_j}{\partial x_i}\right)_{T,v,x_{k\neq i}}\right] \tag{124}$$

$$\left(\frac{\partial \kappa}{\partial x_i}\right)_{T,v,x_{j\neq i}} = \frac{1}{2}\left(\frac{\rho_N e^2}{k_B T \varepsilon_0 \varepsilon_r}\sum_j \alpha_j x_j z_j^2\right)^{-\frac{1}{2}}$$
$$\left\{-\frac{\rho_N e^2}{k_B T\left(\varepsilon_0\varepsilon_r\right)^2}\varepsilon_0\frac{\partial\left(\varepsilon_r\right)}{\partial x_i}\sum_j \alpha_j x_j z_j^2 + \frac{\rho_N e^2}{k_B T \varepsilon_0 \varepsilon_r}\alpha_i z_i^2\right\}$$
$$= \left(\frac{\rho_N e^2}{k_B T}\right)^{\frac{1}{2}}\left\{-\frac{1}{2}\left(\varepsilon_0\varepsilon_r\right)^{-\frac{3}{2}}\varepsilon_0\frac{\partial\left(\varepsilon_r\right)}{\partial x_i}\left[\sum_j \alpha_j x_j z_j^2\right]^{\frac{1}{2}}\right. \tag{125}$$
$$\left. + \frac{1}{2\sqrt{\varepsilon_0\varepsilon_r}}\left[\sum_j \alpha_j x_j z_j^2\right]^{-\frac{1}{2}}\alpha_i z_i^2\right\}$$

$$\left(\frac{\partial \chi_i}{\partial x_i}\right)_{T,v,x_{j\neq i}} = -\frac{9}{\left(\kappa R_i\right)^4}\left[\frac{3}{2} + \ln(1+\kappa R_i) - 2(1+\kappa R_i)\right.$$
$$\left. + \frac{1}{2}\left(1+\kappa R_i\right)^2\right] R_i \frac{\partial \kappa}{\partial x_i} + \frac{3}{\left(\kappa R_i\right)^3}\left[\frac{1}{1+\kappa R_i} - 1 + \kappa R_i\right] \tag{126}$$
$$R_i\frac{\partial \kappa}{\partial x_i} = 3\frac{\partial \kappa}{\partial x_i}\left\{-\frac{\chi_i}{\kappa} + \frac{R_i}{\left(\kappa R_i\right)^3}\left[\frac{1}{1+\kappa R_i} - 1 + \kappa R_i\right]\right\}$$

3.4.2 Dielectric Constant

Mole-Fraction

$$\left(\frac{\partial \varepsilon_r}{\partial x_i}\right)_{x_{j\neq i}} = \varepsilon_{r,i} \tag{127}$$

Mass-Fraction

$$\left(\frac{\partial \varepsilon_r}{\partial x_i}\right)_{x_{j\neq i}} = \frac{MW_i}{\overline{MW}}\left(\varepsilon_{r,i} - \varepsilon_r\right) \tag{128}$$

Combo

$$\left(\frac{\partial \varepsilon_r}{\partial x_i}\right)_{x_{j\neq i}} = \varepsilon_{r,sol}^{(w)} + x_{sol}\,\frac{MW_i}{\overline{MW}_{sol}}\left(\varepsilon_{r,i} - \varepsilon_{r,sol}^{(w)}\right), \qquad i \in \mathcal{S} \tag{129}$$

New Mixing Rule

$$\varepsilon_{r,\mathrm{mix}}(\mathbf{x}) = \frac{\varepsilon_{sf}(\mathbf{x})}{1 + 7.01\, x_{\mathrm{ion}}(\mathbf{x})}, \qquad \varepsilon_{sf} \equiv \varepsilon_{r,\mathrm{solvent,mix}}^{\mathrm{salt\text{-}free}}. \tag{130}$$

$$\left(\frac{\partial \varepsilon_{r,\mathrm{mix}}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = \frac{1}{1 + 7.01\, x_{\mathrm{ion}}}\left(\frac{\partial \varepsilon_{sf}}{\partial x_i}\right)_{T,v,x_{j\neq i}} - \frac{7.01\, \varepsilon_{sf}}{\left(1 + 7.01\, x_{\mathrm{ion}}\right)^2}\left(\frac{\partial x_{\mathrm{ion}}}{\partial x_i}\right)_{T,v,x_{j\neq i}}. \tag{131}$$

$$\varepsilon_{sf}(\mathbf{x}) = \sum_{s\in\mathcal{S}} w_s^{sf}\, \varepsilon_{r,s}, \qquad w_s^{sf} = \frac{x_s M_s}{\sum_{m\in\mathcal{S}} x_m M_m}, \qquad D \equiv \sum_{m\in\mathcal{S}} x_m M_m. \tag{132}$$

$$\left(\frac{\partial \varepsilon_{sf}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = \sum_{s\in\mathcal{S}} \varepsilon_{r,s}\left(\frac{\partial w_s^{sf}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = \begin{cases} \dfrac{M_i}{D}\left(\varepsilon_{r,i} - \varepsilon_{sf}\right), & i \in \mathcal{S}, \\ 0, & i \notin \mathcal{S}. \end{cases} \tag{133}$$

$$x_{\mathrm{ion}}(\mathbf{x}) = \sum_{m\in\mathcal{I}} x_m, \qquad \left(\frac{\partial x_{\mathrm{ion}}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = \begin{cases} 1, & i \in \mathcal{I}, \\ 0, & i \notin \mathcal{I}. \end{cases} \tag{134}$$

$$\left(\frac{\partial \varepsilon_{r,\mathrm{mix}}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = \begin{cases} \dfrac{1}{1 + 7.01\, x_{\mathrm{ion}}}\,\dfrac{M_i}{\sum_{m\in\mathcal{S}} x_m M_m}\left(\varepsilon_{r,i} - \varepsilon_{sf}\right), & i \in \mathcal{S}, \\ -\dfrac{7.01\, \varepsilon_{sf}}{\left(1 + 7.01\, x_{\mathrm{ion}}\right)^2}, & i \in \mathcal{I}. \end{cases} \tag{135}$$

3.5 Born Electrolyte Term Contribution

$$\tilde{\mu}_k^{Born} = \tilde{a}^{\mathrm{Born}} + Z^{Born} + \left(\frac{\partial \tilde{a}^{\mathrm{Born}}}{\partial x_k}\right)_{T,\nu,x_{i\neq k}} - \sum_{j=1}^{N}\left[x_j\left(\frac{\partial \tilde{a}^{\mathrm{Born}}}{\partial x_j}\right)_{T,\nu,x_{i\neq j}}\right] \tag{136}$$

Version 1: How the reference "ePC-SAFT advanced, Part I: Physical meaning of including a concentration-dependent dielectric constant in the born term and in the Debye-Hückel theory" has it (equations 3) [4]

$$\left(\frac{\partial \tilde{a}^{Born}}{\partial x_i}\right)_{T,v_N,x_{j\neq i}} = -\frac{e^2}{4\pi k_B T \varepsilon_0}\left[\left(1-\frac{1}{\varepsilon_r}\right)\frac{z_i^2}{d_i^{\mathrm{Born}}} + \left(\frac{1}{\varepsilon_r^2}\right)\left(\frac{\partial \varepsilon_r}{\partial x_i}\right)_{T,v,x_{j\neq i}}\right] \tag{137}$$

Version 2: Potentially Fixed Differential from the paper

$$\left(\frac{\partial \tilde{a}^{Born}}{\partial x_i}\right)_{T,v_N,x_{j\neq i}} = -\frac{e^2}{4\pi k_B T \varepsilon_0}\left[\left(1-\frac{1}{\varepsilon_r}\right)\frac{z_i^2}{d_i^{\mathrm{Born}}} + \left(\frac{1}{\varepsilon_r^2}\right)\sum_j \frac{x_j z_j^2}{d_j^{\mathrm{Born}}}\left(\frac{\partial \varepsilon_r}{\partial x_i}\right)_{T,v,x_{j\neq i}}\right] \tag{138}$$

3.5.1 Solvation Shell Model + Dielectric Saturation

See [7]. Version 3

$$\begin{aligned}\left(\frac{\partial \tilde{a}^{\mathrm{Born}}_{\mathrm{SSM+DS}}}{\partial x_k}\right)_{T,v,x_{j\neq k}} = -\frac{e^2}{4\pi\varepsilon_0 k_{\mathrm{B}} T}\Bigg[&\left(1-\frac{1}{\varepsilon_r}\right)\frac{z_k^2}{d_k^{\mathrm{Born}}+\Delta d_k} \\ &+\left(1-\frac{1}{\varepsilon_{r,\mathrm{ion}}}\right) z_k^2\left(\frac{1}{d_k^{\mathrm{Born}}} - \frac{1}{d_k^{\mathrm{Born}}+\Delta d_k}\right) \\ &+\left(\frac{1}{\varepsilon_r^2}\right)\left(\frac{\partial \varepsilon_r}{\partial x_k}\right)_{T,v,x_{j\neq k}}\left(\sum_i x_i z_i^2 \frac{1}{d_i^{\mathrm{Born}}+\Delta d_i}\right)\Bigg].\end{aligned} \tag{139}$$

Version 4

$$\left(\frac{\partial a^{\mathrm{Born}}}{\partial x_i}\right)_{T,v,x_{j\neq i}} = -\frac{e^2}{4\pi\varepsilon_0 k_B T}\Bigg[z_i^2\Bigg(\left(\left(1-\frac{1}{\varepsilon_{r,\mathrm{ion}}}\right)\frac{1}{D_i}+\left(1-\frac{1}{\varepsilon_{r,\mathrm{solv}}}\right)\left(\frac{1}{d_i^{\mathrm{Born}}}-\frac{1}{D_i}\right)\right)$$

$$+\sum_j x_j z_j^2\Bigg(\left(\left(\frac{1}{\varepsilon_{r,\mathrm{ion}}}-\frac{1}{\varepsilon_{r,\mathrm{solv}}}\right)\frac{1}{D_j^2}\frac{\partial \Delta d_m}{\partial x_i}\right.$$

$$\left.+\frac{1}{\varepsilon_{r,\mathrm{solv}}^2}\frac{\partial \varepsilon_{r,\mathrm{solv}}}{\partial x_i}\left(\frac{1}{d_j^{\mathrm{Born}}}-\frac{1}{D_j}\right)\right)\Bigg)\Bigg], \tag{140}$$

$$\frac{\partial \Delta d_m}{\partial x_i} = \frac{d_m^{\mathrm{Born}}}{|z_m|}\frac{\partial f_{\mathrm{mix}}}{\partial x_i} = \frac{d_m^{\mathrm{Born}}}{|z_m|} f_i, \tag{141}$$

$$D_m = d_m^{\mathrm{Born}} + \Delta d_m, \Delta d_m = \frac{(f_{\mathrm{mix}}-1)}{|z_m|} d_m^{\mathrm{Born}}\ . \tag{142}$$

4 Fugacity Coefficient

See [8].

$$\ln \varphi_k = \tilde{\mu}_k^{\mathrm{res}} - \ln Z = \frac{\mu_k^{\mathrm{res}}}{kT} - \ln Z = \sum_\alpha \ln \varphi^\alpha = \sum_\alpha \mu^\alpha + \sum_\alpha \left[-\frac{Z^\alpha}{Z-1}\ln Z\right] \tag{143}$$

$$\ln \varphi_k = \ln \varphi_k^{hc} + \ln \varphi_k^{disp} + \ln \varphi_k^{assoc} + \ln \varphi_k^{DH} + \ln \varphi_k^{Born} \tag{144}$$

$$\ln \varphi_k^\alpha = \tilde{\mu}_k^\alpha - \frac{Z^\alpha}{Z-1}\ln Z \tag{145}$$

$$\ln \varphi_k^\alpha \approx \tilde{\mu}_k^\alpha - Z^\alpha \tag{146}$$

4.1 Hard-Chain Reference Contribution

$$\ln \varphi_k^{hc} = \tilde{\mu}_k^{hc} - \frac{Z^{hc}}{Z-1} \ln Z \tag{147}$$

4.2 Dispersion Contribution

$$\ln \varphi_k^{disp} = \tilde{\mu}_k^{disp} - \frac{Z^{disp}}{Z-1} \ln Z \tag{148}$$

4.3 Association Contribution

$$\ln \varphi_k^{assoc} = \tilde{\mu}_k^{assoc} - \frac{Z^{assoc}}{Z-1} \ln Z \tag{149}$$

4.4 Debye and Huckel Electrolyte Term Contribution

$$\ln \varphi_k^{DH} = \tilde{\mu}_k^{DH} - \frac{Z^{DH}}{Z-1} \ln Z \tag{150}$$

4.5 Born Electrolyte Term Contribution

$$\ln \varphi_k^{Born} = \tilde{\mu}_k^{Born} - \frac{Z^{Born}}{Z-1} \ln Z \tag{151}$$

5 Activity Coefficient

See [8, 7].

$$\gamma_i = \frac{\varphi_i(T, p, \mathbf{x})}{\varphi_{0i}(T, p, x_i = 1)} \tag{152}$$

$$\gamma_i^* = \frac{\varphi_i(T, p, \mathbf{x})}{\varphi_i^\infty(T, p, x_i \to 0)} \tag{153}$$

$$\ln \gamma_i = \ln \varphi_i - \ln \varphi_{0i} \tag{154}$$

$$\ln \gamma_i^* = \ln \varphi_i - \ln \varphi_i^{\infty} \tag{155}$$

$$\ln \gamma_k^* = \ln \gamma_k^{hc,*} + \ln \gamma_k^{disp,*} + \ln \gamma_k^{assoc,*} + \ln \gamma_k^{DH,*} + \ln \gamma_k^{Born,*} \tag{156}$$

$$\gamma_k^{\alpha,*} = \frac{\varphi_k^{\alpha}}{\varphi_k^{\alpha,\infty}} \tag{157}$$

$$\ln \gamma_k^{\alpha,*} = \ln \varphi_k^{\alpha} - \ln \varphi_k^{\alpha,\infty} \tag{158}$$

$$\gamma_{\pm}^* = \left((\gamma_c^*)^{\nu_c} (\gamma_a^*)^{\nu_a} \right)^{\frac{1}{\nu_c+\nu_a}} \tag{159}$$

$$\ln \gamma_{\pm}^* = \frac{\nu_c \ln \gamma_c^* + \nu_a \ln \gamma_a^*}{\nu_c + \nu_a} \tag{160}$$

$$\ln \gamma_{\pm}^{\alpha,*} = \frac{\nu_c \ln \gamma_c^{\alpha,*} + \nu_a \ln \gamma_a^{\alpha,*}}{\nu_c + \nu_a} \tag{161}$$

5.1 Hard-Chain Reference Contribution

$$\ln \gamma_k^{hc,*} = \ln \varphi_k^{hc} - \ln \varphi_k^{hc,\infty} \tag{162}$$

5.2 Dispersion Contribution

$$\ln \gamma_k^{disp,*} = \ln \varphi_k^{disp} - \ln \varphi_k^{disp,\infty} \tag{163}$$

5.3 Association Contribution

$$\ln \gamma_k^{assoc,*} = \ln \varphi_k^{assoc} - \ln \varphi_k^{assoc,\infty} \tag{164}$$

5.4 Debye and Huckel Electrolyte Term Contribution

$$\ln \gamma_k^{DH,*} = \ln \varphi_k^{DH} - \ln \varphi_k^{DH,\infty} \tag{165}$$

5.5 Born Electrolyte Term Contribution

$$\ln \gamma_k^{Born,*} = \ln \varphi_k^{Born} - \ln \varphi_k^{Born,\infty} \tag{166}$$

6 Enthalpy, Entropy, and Gibbs Free Energy

6.1 Entropy

$$\tilde{h}^{\mathrm{res}} = \frac{\hat{h}^{\mathrm{res}}}{RT} = -T\left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial T}\right)_{\rho,x_i} + (Z-1) \tag{167}$$

6.2 Entropy

$$\tilde{s}^{\mathrm{res}} = \frac{\hat{s}^{\mathrm{res}}(P,T)}{R} = \frac{\hat{s}^{\mathrm{res}}(\nu,T)}{R} + \ln(Z) \tag{168}$$

$$\tilde{s}^{\mathrm{res}} = \frac{\hat{s}^{\mathrm{res}}(P,T)}{R} = -T\left[\left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial T}\right)_{\rho,x_i} + \frac{\tilde{a}^{\mathrm{res}}}{T}\right] + \ln(Z) \tag{169}$$

6.3 Gibbs Free Energy

$$\tilde{g}^{\mathrm{res}} = \frac{\hat{g}^{\mathrm{res}}}{RT} = \frac{\hat{h}^{\mathrm{res}}}{RT} - \frac{\hat{s}^{\mathrm{res}}(P,T)}{R} \tag{170}$$

$$\tilde{g}^{\mathrm{res}} = \tilde{a}^{\mathrm{res}} + (Z-1) - \ln(Z) \tag{171}$$

Gibbs Energy of Solvation

$$\Delta^{\mathrm{solv},x} G_i^{\infty}(T,p,x_{j\neq i},x_i \to 0) = RT\ln(\varphi_i^{\infty}(T,p,x_{j\neq i},x_i \to 0)) \tag{172}$$

Gibbs Energy of Transfer at Infinite Dilution from solvent S1 to solvent S2

$$\Delta^{\mathrm{tr}} G_i^{\infty,\mathrm{S1}\rightarrow\mathrm{S2}}(T, p, x_{j\neq i}, x_i \rightarrow 0) = RT \ln\left(\frac{\varphi_i^{\infty,\mathrm{S2}}}{\varphi_i^{\infty,\mathrm{S1}}}\right) \tag{173}$$

6.4 Temperature Differential Equations

$$\left(\frac{\partial \tilde{a}^{\mathrm{res}}}{\partial T}\right)_{\rho,x_i} = \left(\frac{\partial \tilde{a}^{\mathrm{hc}}}{\partial T}\right)_{\rho,x_i} + \left(\frac{\partial \tilde{a}^{\mathrm{disp}}}{\partial T}\right)_{\rho,x_i} + \left(\frac{\partial \tilde{a}^{\mathrm{assoc}}}{\partial T}\right)_{\rho,x_i} + \left(\frac{\partial \tilde{a}^{\mathrm{DH}}}{\partial T}\right)_{\rho,x_i} + \left(\frac{\partial \tilde{a}^{\mathrm{Born}}}{\partial T}\right)_{\rho,x_i} \tag{174}$$

6.4.1 Hard-Chain Reference Contribution

$$\left(\frac{\partial \tilde{a}^{\mathrm{hc}}}{\partial T}\right)_{\rho,x_i} = \bar{m}\left(\frac{\partial \tilde{a}^{\mathrm{hs}}}{\partial T}\right)_{\rho,x_i} - \sum_i x_{\mathrm{i}}(m_{\mathrm{i}} - 1)(g_{ii}^{\mathrm{hs}})^{-1}\left(\frac{\partial g_{ii}^{\mathrm{hs}}}{\partial T}\right)_{\rho,x_i} \tag{175}$$

$$\begin{aligned}\left(\frac{\partial \tilde{a}^{hs}}{\partial T}\right)_{\rho,x_i} = \frac{1}{\zeta_0}\Bigg[&\frac{3(\zeta_{1,T}\zeta_2 + \zeta_1\zeta_{2,T})}{(1-\zeta_3)} + \frac{3\zeta_1\zeta_2\zeta_{3,T}}{(1-\zeta_3)^2} \\ &+ \frac{3{\zeta_2}^2\zeta_{2,T}}{\zeta_3(1-\zeta_3)^2} + \frac{{\zeta_2}^3\zeta_{3,T}(3\zeta_3 - 1)}{{\zeta_3}^2(1-\zeta_3)^3} \\ &+ \left(\frac{3{\zeta_2}^2\zeta_{2,T}\zeta_3 - 2{\zeta_2}^3\zeta_{3,T}}{{\zeta_3}^3}\right)\ln(1-\zeta_3) + \left(\zeta_0 - \frac{{\zeta_2}^3}{{\zeta_3}^2}\right)\frac{\zeta_{3,T}}{(1-\zeta_3)}\Bigg]\end{aligned} \tag{176}$$

$$\begin{aligned}\frac{\partial g_{ii}^{\mathrm{hs}}}{\partial T} = &\frac{\zeta_{3,T}}{(1-\zeta_3)^2} + \left(\frac{1}{2}d_{i,T}\right)\frac{3\zeta_2}{(1-\zeta_3)^2} \\ &+ \left(\frac{1}{2}d_i\right)\left(\frac{3\zeta_{2,T}}{(1-\zeta_3)^2} + \frac{6\zeta_2\zeta_{3,T}}{(1-\zeta_3)^3}\right) + \left(\frac{1}{2}d_i d_{i,T}\right)\frac{2\zeta_2^2}{(1-\zeta_3)^3} \\ &+ \left(\frac{1}{2}d_i\right)^2\left(\frac{4\xi_2\xi_{2,T}}{(1-\xi_3)^3} + \frac{6{\xi_2}^2\xi_{3,T}}{(1-\xi_3)^4}\right)\end{aligned} \tag{177}$$

$$d_{i,T} = \frac{\partial d_i}{\partial T} = \sigma_i\left(3\frac{\epsilon_i}{kT^2}\right)\left[-0.12\exp\left(-3\frac{\epsilon_i}{kT}\right)\right] \tag{178}$$

$$\zeta_{n,T} = \frac{\partial \zeta_n}{\partial T} = \frac{\pi}{6}\rho\sum_i x_i m_i n d_{i,T}\,(d_i)^{n-1} \quad n \in \{1,2,3\} \tag{179}$$

$$\frac{1}{2}d_i = \left(\frac{d_i d_i}{d_i + d_i}\right) \tag{180}$$

6.4.2 Dispersion Contribution

$$\left(\frac{\partial \tilde{a}^{\mathrm{disp}}}{\partial T}\right)_{\rho,x_i} = -2\pi\rho\left(\frac{\partial I_1}{\partial T} - \frac{I_1}{T}\right)\overline{m^2\epsilon\sigma^3} - \pi\rho\bar{m}\left[\frac{\partial C_1}{\partial T}I_2 + C_1\frac{\partial I_2}{\partial T} - 2C_1\frac{I_2}{T}\right]\overline{m^2\epsilon^2\sigma^3} \tag{181}$$

6.4.3 Debye and Huckel Electrolyte Term Contribution

$$\frac{d\tilde{a}^{DH}}{dT} = -\frac{1}{12\pi k_B\varepsilon_0\varepsilon_r}\sum_i x_i\left(z_i e\right)^2\left[\left(-2\chi_i + \frac{3}{1+\kappa a_i}\right)\left(-\frac{\kappa}{2T^2}\right) - \frac{\kappa\chi_i}{T^2}\right] \tag{182}$$

6.4.4 Born Contribution

$$\left(\frac{\partial \tilde{a}^{\mathrm{Born}}}{\partial T}\right)_{\rho,x_i} = \frac{e^2}{4\pi\varepsilon_0 k_B T^2}\left(1 - \frac{1}{\varepsilon_r}\right)\sum_i \frac{x_i z_i^2}{a_i} \tag{183}$$

7 Salt Basis Conversions

$$\gamma_{\pm}^{*,m} = \frac{\gamma_{\pm}^{*,x}}{1 + M_{\mathrm{solvent}} \cdot \tilde{m}_{\mathrm{solute}} \cdot \sum_i \nu_{i,\mathrm{ion}}} \tag{184}$$

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